

# Numbers and Sets

ECON 441: Introduction to Mathematical Economics  
Lecture 1

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## Introductions

- Your preferred name and pronouns
- Why are you here?
  - Undergrads: why you signed up for this class
  - Grad students: why you're doing the MA
- The last math class you took, and roughly when
- Your comfort food

## Real-Number System

- *Integers:*

$\dots, -3, -2, -1, 0, 1, 2, 3, \dots$

- *Fractions:*

$$\frac{1}{2}, \frac{3}{5}, -\frac{2}{3}$$

- *Rational numbers:* ratio of integers
- *Are fractions rational numbers? What about integers?*

## Real-Number System

- *Rational numbers*: ratio of integers
  - “terminating or repeating decimal”
  - Example.  $\frac{1}{3} = 0.333$ ,  $\frac{1}{4} = 0.25$
- *Irrational numbers*: cannot be expressed as a ratio of two integers
  - “nonterminating or nonrepeating decimal”
  - Example.  $\sqrt{2} = 1.4142$ ,  $\pi = 3.1415$
- *Real numbers* ( $\mathbb{R}$ ): rational and irrational

## Sets

- A *set* is a collection of distinct objects.

$$A = \{brownies, icecream, pizza, ramen\}$$

- $pizza \in A$ ,  $\in$  stands for 'is in'
- Other ways of writing a set

$$B = \{x | x \text{ is a positive integer}\}$$

$$C = \{x | 1 < x < 5\}$$

## Set Relations

- Equivalence ( $=$ )

$$A = \{brownies, icecream, pizza, ramen\}$$

$$B = \{pizza, ramen, icecream, brownies\}$$

$$A = B$$

- Subset ( $\subset$ )

$$C = \{pizza, ramen\}$$

$C \neq A$  but  $C \subset A$ . Note:  $A \supset C$  is equivalent.

Is  $A \subset B$ ?

## Set Relations

- Disjoint sets

$$A = \{brownies, icecream, pizza, ramen\}$$

$$D = \{salad, fruits\}$$

- Neither but still related

$$A = \{brownies, icecream, pizza, ramen\}$$

$$E = \{salad, fruits, icecream\}$$

## Set Relations

- $\emptyset$ : empty or null set
- What are all possible subsets of  $S = \{a, b, c\}$ ?

$\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a, b, c\}$

Always  $2^n$  subsets. Here  $n = 3$ , so 8 subsets.

## Set Operations

- *Union:  $A \cup B$ , elements in either  $A$  or  $B$*
- *Intersection:  $A \cap B$ , elements in both  $A$  and  $B$*

Example:

$$A = \{brownies, icecream, pizza, ramen\}$$

$$B = \{salad, fruits, icecream\}$$

$$A \cup B =$$

$$A \cap B =$$

## Set Operations

- *Union*:  $A \cup B$ , elements in either  $A$  or  $B$
- *Intersection*:  $A \cap B$ , elements in both  $A$  and  $B$

What about

$$A = \{brownies, icecream, pizza, ramen\}$$

$$B = \{salad, fruits\}$$

$$A \cup B =$$

$$A \cap B =$$

## Set Operations

- *Complement of A*:  $\tilde{A}$  or  $A^c$ , 'not  $A$ '
- Universal set  $U$  (context specific) then:

$$A^c = \{x | x \in U \text{ and } x \notin A\}$$

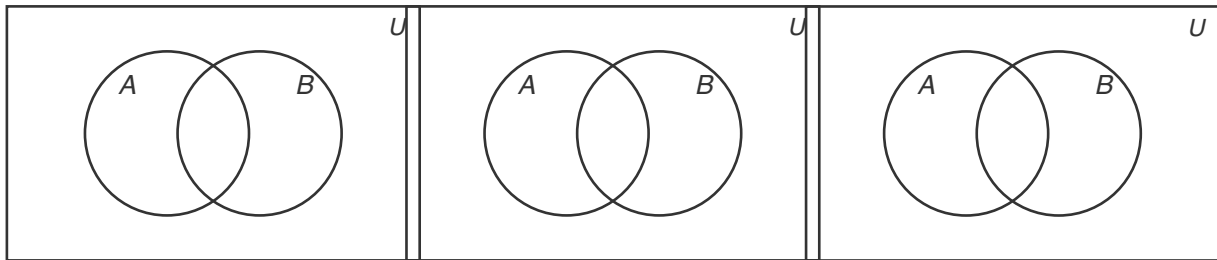
Example.  $U = \{1, e, f, 2\}$ ,  $A = \{1, 2\}$ , then  $A^c = \{e, f\}$ .

Set Operations: Venn Diagrams    Shade the following regions:

$A \cup B$

$A \cap B$

$A^c$



## Laws of Set Operations

- Commutative law

$$A \cup B = B \cup A \quad A \cap B = B \cap A$$

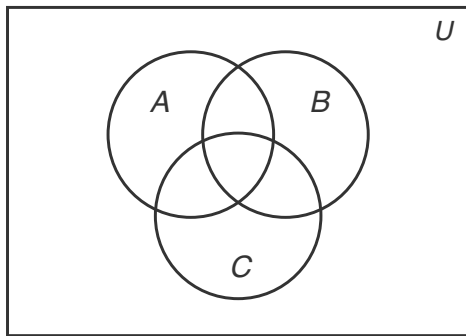
- Distributive law

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

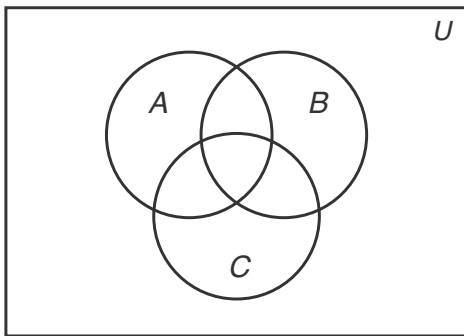
$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Distributive law Verify by shading:  $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

$A$

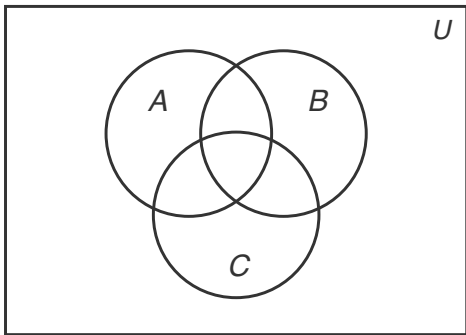


$B \cap C$

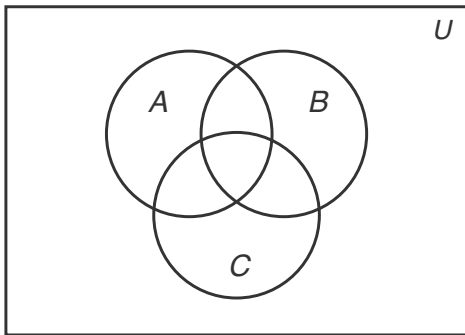


Distributive law Now the right-hand side,  $(A \cup B) \cap (A \cup C)$ :

$A \cup B$



$A \cup C$



## Ordered Sets

- We said order does not matter for sets
- But we can have ordered sets where

$$(a, b) \neq (b, a) \text{ unless } a = b$$

- Ordered pairs, triples,...

Example.  $(age, weight)$ ,  $(22, 120)$  different from  $(120, 22)$

Cartesian Product

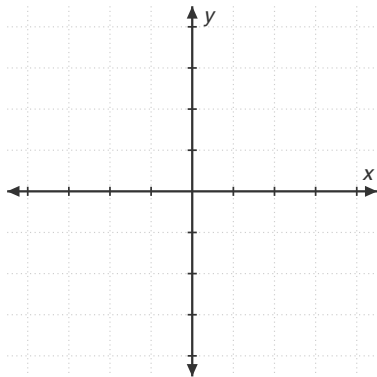
$$A = \{1, 2\} \quad B = \{3, 4\}$$

Cartesian Product: set of all possible ordered pairs

$$A \times B = \{(1, 3), (1, 4), (2, 3), (2, 4)\}$$

Cartesian Plane

$$\mathbb{R}^2 = \{(x, y) | x \in \mathbb{R}, y \in \mathbb{R}\}$$



Can have  $\mathbb{R}^3, \mathbb{R}^4, \dots, \mathbb{R}^n$

By the way What is  $x \times x$ ?

What is  $x^2 \times x$ ?

What is  $x^2 \times x^2$ ?

More generally,  $x^n \times x^m = x^{m+n}$